Imdadul Haque Sourav

imdadulhaquesourav@gmail.com | Dhaka, Bangladesh | GitHub | LinkedIn

Research Interests

Deep Learning, Medical Image Analysis, Multimodal and Explainable AI, Foundation Models.

Education

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
Bachelor of Science in Biomedical Engineering CGPA: 3.61/4.00
(3.69 upper-division)

Experience

Remote Research Intern (Part-time) June 2026 – Present
PI: Dr. Sayeed Shafayeet Chowdhury, Department of Computer Science, Indiana University Indianapolis

- Collaborated with the Department of Biology and Neuroscience at IU Indianapolis to acquire and analyze rodent retinal imaging data.
- Developed a transfer learning pipeline for predicting optic nerve head morphometric measurements from en-face SD-OCT images of rodent retinas.
- Benchmarked frozen and fine-tuned vision foundation models (DINOv2, RETFound, BiomedCLIP, EVA-02) against classical computer vision baselines with animal-wise cross-validation.

Intern (Biomedical Engineering) March 9–24, 2025
Global Medical Engineering (BD) Ltd. Dhaka, Bangladesh

- Operated and analyzed medical devices, including ultrasound machines, diagnostic analyzers, ECG, and infusion pumps.
- Performed sample testing, parameter calculation, and image interpretation; wrote technical reports.

Projects

Epidemiologically-Informed Guided Attention for Oral Cavity Lesion Classification from Smartphone Imagery
Undergraduate Thesis | Supervisor: Dr. Taufiq Hasan, Professor, BME, BUET January 2025 – Present
Presentation | Thesis book

- Developed a multitask deep learning framework with dual classification heads and Guided Attention Pooling to segment and classify oral lesions.
- Designed a novel Epidemiological Prior Layer that integrates patient risk factors into the neural network, combining visual features with clinical context.
- Led the curation of the first clinical oral cancer dataset in Bangladesh (500 images across 130 patients) in partnership with the Shanti Cancer Foundation, Dhaka.

Development of an Affordable Video Laryngoscope for Resource-Constrained Settings July 2025
PI: Abdullah Al Mamun, Lecturer, BME, BUET Presentation

- Engineered a 3D-printed video laryngoscope, achieving an 80% cost reduction (from \$1,200+ to \$35–\$70).
- Validated performance through mannequin intubation trials and endoscopic video capture in collaboration with BUET and Asgar Ali Hospital, Dhaka.

Consensus-Driven Label Cleaning for Robust Fundus Disease Classification June 2026 – Present
Independent Project Presentation

- Designed a consensus-driven label cleaning pipeline using multi-model agreement to identify and remove annotation errors from a noisy clinical fundus dataset, improving training label quality beyond what single-model filtering can achieve.
- Proposed two custom feature-level loss functions: a Frequency-Adaptive Anchor Loss that applies stronger pulls on rare-class samples in embedding space, and a Confusion-Aware Margin Loss that separates clinically confusable class pairs, addressing class imbalance and inter-class confusion simultaneously.

Impact of Different Image Processing Techniques on Retina Lesions Classification July 2025
PI: Samiul Based Shuvo, Assistant Professor, BME, BUET Presentation

- Trained and evaluated five deep learning architectures to classify four retinal conditions on a dataset of over 32,000 optical coherence tomography (OCT) scans.
- Boosted model accuracy by up to 4.75% through a custom preprocessing pipeline that removed invalid image borders and applied adaptive contrast enhancement (CLAHE) to highlight abnormalities.

Ensemble-Based Deep Learning System for Diabetic Retinopathy Detection January 2025
PI: Shams Nafisa Ali, Assistant Professor, BME, BUET GitHub

- Developed an ensemble deep learning model to classify diabetic retinopathy from retinal images on the APTOS dataset.
- Deployed an interactive web app with real-time prediction using a max-voting ensemble strategy.

Fingerprint Recognition System for Secure Patient Data Access January 2025
PI: Dr. Md. Kamrul Hasan, Professor, EEE, BUET Presentation

- Developed a complete hardware-software system using digital signal processing for image enhancement, feature extraction, and matching.
- Integrated biometric authentication with patient data access, enabling automatic retrieval of medical files.

Simulation and Comparison of Nonlinear Pressure Controllers for Soft Pneumatic Actuators
PI: Shams Nafisa Ali, Assistant Professor, BME, BUET Presentation

- Implemented and replicated Feedback Linearization, Sliding Mode, and PI controllers in MATLAB/Simulink based on published research.
- Analyzed and compared control performance with the original paper's results to evaluate accuracy and practical applicability.

Honors and Awards

Semi-finalist – Rice360 Global Health Technologies Design Competition April 2025
Led the team and presented the video laryngoscope prototype in person at **Rice University, Houston, TX**.

Selected Presenter – Carnegie Mellon Forum on Biomedical Engineering September 2025
Presented on the design and development of an affordable video laryngoscope at **Carnegie Mellon University**.

1st Place – Project Showcasing Competition, 1st National Biomedical Conference, JUST June 2025
Presented a biomedical engineering project, securing first place among competing teams at Jessore University of Science and Technology.

Accepted Abstract – World Cancer Congress Hong Kong, September 2026
Undergraduate thesis accepted as an ePoster in the abstract eLibrary of the World Cancer Congress.

Skills and Proficiencies

Programming Languages: Python, MATLAB, C
ML Frameworks: TensorFlow/Keras, PyTorch
Python Libraries: NumPy, pandas, Matplotlib, scikit-learn, OpenCV, SciPy, Streamlit
Embedded Systems & Hardware: ATmega32 Microcontroller, Arduino, Raspberry Pi
CAD & Simulations: SOLIDWORKS, Ansys, COMSOL Multiphysics, Proteus

Relevant Coursework

Computer Programming, Digital Signal Processing, Linear Algebra, Medical Imaging, Probability & Statistics, Bioinformatics, Embedded Systems and Interfacing, Machine Learning for Biomedical Engineers.